BB_22_Vs_23

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```r
# Load libraries
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.4.4     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.0
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```r
#read in the data
BB_22_23_fit <- read_csv(here::here("data_sheets", "compiled_sheets", "BB_22_23_fit.csv"))
```

```
## New names:
## Rows: 870 Columns: 74
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
```

```
## (8): Gen, Donor, Recipient, sample_ID, SegPos, sample_ID_SEG, Replicat... dbl
## (50): Year, Location, Sequence, Plant_ID, days_sow2flower, days_plant2f... lgl
## (6): F_plant, Rep_FitP, any_FitP, Germ_Date, days_germ2flower, ...70 date (10):
## Sow_Date, Plant_Date, FFD, LFD, FD_06, FD_07, FD_08, FD_09, photo...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `...36` -> `...70`
```

```r
# Create a new column "Genotype" by merging "Donor" and "Recipient" with a hyphen in between
BB_22_23_fit$Genotype <- paste(BB_22_23_fit$Donor, BB_22_23_fit$Recipient, sep = "-")

BB_22_23_full <- read_csv(here::here("data_sheets", "compiled_sheets", "BB_22_23_full.csv"))
```

```
## New names:
## Rows: 7073 Columns: 74
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (9): Gen, Donor, Recipient, sample_ID, SegPos, sample_ID_SEG, Block, R... dbl
## (50): Year, Location, Sequence, Plant_ID, days_sow2flower, days_plant2f... lgl
## (4): F_plant, Rep_FitP, any_FitP, ...70 date (11): Sow_Date, Plant_Date, FFD,
## LFD, FD_06, FD_07, FD_08, FD_09, photo...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `...36` -> `...70`
```

```r
# Calculate the mean surv_to_flower for each group in AC_22_23_full and add it as a new column
BB_22_23_full <- BB_22_23_full %>%
  group_by(Year, Gen, Location, Sequence, Donor, Recipient, SegPos) %>%
  mutate(prop_surv_to_flower = mean(surv_to_flower, na.rm = TRUE)) %>%
  ungroup()  # Remove grouping

BB_prop_sample <- BB_22_23_full %>%
  select(c(Year, Gen, Location, Sequence, Donor, Recipient, SegPos, prop_surv_to_flower))  %>%
    distinct()

BB_22_23_fit <- BB_22_23_fit %>%
  left_join(BB_prop_sample %>% select(Year, Gen, Location, Sequence, Donor, Recipient, SegPos, prop_surv
            by = c("Year", "Gen", "Location", "Sequence", "Donor", "Recipient", "SegPos"))

# Create the new column
BB_22_23_fit <- BB_22_23_fit %>%
  mutate(est_fitness = prop_surv_to_flower * est_fecundity)
```

```r
#Subset the data for eBBh year and eBBh generation
BB_22_fit <- BB_22_23_fit %>%
  filter(Year == 2022)

BB_23_fit_G1 <- BB_22_23_fit %>%
  filter(Year == 2023) %>%
  filter(Gen == "G1")

BB_23_fit_G2 <- BB_22_23_fit %>%
  filter(Year == 2023) %>%
  filter(Gen == "G2")
```

Now compare Generation 1 from BB in both 2022 and 2023, but first we need to determine which genotypes

are the same in 2022 and 2023

```
# Create a dataframe with common "Genotype" values for 2022
BB_common_22_23_G1_22 <- semi_join(BB_22_fit, BB_23_fit_G1, by = "Genotype")

# Create a dataframe with common "Genotype" values for 2022
BB_common_22_23_G1_23 <- semi_join(BB_23_fit_G1, BB_22_fit, by = "Genotype")

# Merge these two dataframes to have one finalized dataset to compare G1 2022 and G1 2023
BB_common_22_23_G1 <- full_join(BB_common_22_23_G1_22, BB_common_22_23_G1_23, by = intersect(names(BB_c
```

Determine if 2022 and 2023 trait means are significantly different in Generation 1 Any difference would be due to environmental effects (because these are the same gneotypes)

#MATERNAL#2022v2023#

```
# Define the traits to analyze
traits <- c("corolla_diam_mm", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG", "fl

# Loop over eBBh trait
for (trait in traits) {

  # Calculate the mean trait value for eBBh maternal (Recipient) family for 2022
  maternal_means_22 <- BB_common_22_23_G1_22 %>%
    group_by(Recipient, Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE))


  # Calculate the mean trait value for eBBh maternal (Recipient) family for 2023 G1
  maternal_means_23_G1 <- BB_common_22_23_G1_23 %>%
    group_by(Recipient, Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE))


  # Add year information
  maternal_means_22 <- maternal_means_22 %>%
    mutate(Year = "2022")

  maternal_means_23_G1 <- maternal_means_23_G1 %>%
    mutate(Year = "2023_G1")

  # Combine the dataframes
  mat_combined_data_year <- bind_rows(maternal_means_22, maternal_means_23_G1)

  # Create the histogram
  ggplot(mat_combined_data_year, aes(x = mean_trait, fill = Year)) +
    geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
    labs(title = paste("Comparison of Recipient Mean", trait),
         x = paste("Mean", trait),
         y = "Frequency") +
    scale_fill_manual(values = c("2022" = "blue", "2023_G1" = "red")) +
    theme_minimal()

  # Split the data by year
  trait_2022 <- na.omit(maternal_means_22$mean_trait)
  trait_2023 <- na.omit(maternal_means_23_G1$mean_trait)
```

```r
# Check normality with Shapiro-Wilk test
shapiro_test_2022 <- shapiro.test(trait_2022)
shapiro_test_2023 <- shapiro.test(trait_2023)

# Print Shapiro-Wilk test results
print(paste("Shapiro-Wilk test for 2022", trait, ":"))
print(shapiro_test_2022)

print(paste("Shapiro-Wilk test for 2023", trait, ":"))
print(shapiro_test_2023)

# Perform Levene's test for homogeneity of variances
levene_test <- leveneTest(mean_trait ~ Year, data = mat_combined_data_year)
print(paste("Levene's test for", trait, ":"))
print(levene_test)

# Perform Mann-Whitney U test (Wilcoxon rank-sum test)
mann_whitney_test <- wilcox.test(mean_trait ~ Year, data = mat_combined_data_year)
print(paste("Mann-Whitney U test for", trait, ":"))
print(mann_whitney_test)

# Optionally perform t-test if data is normally distributed
# if (shapiro_test_2022$p.value > 0.05 && shapiro_test_2023$p.value > 0.05 && levene_test$`Pr(>F)`[1] 
  t_test_result_year <- t.test(mean_trait ~ Year, data = mat_combined_data_year)
  print(paste("t-test for", trait, ":"))
  print(t_test_result_year)
#} else {
#  print(paste("Skipping t-test for", trait, "due to non-normal distribution or unequal variances"))
}
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
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## `.groups` argument.
```

```
## [1] "Shapiro-Wilk test for 2022 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.99321, p-value = 0.8021
##
## [1] "Shapiro-Wilk test for 2023 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.96694, p-value = 0.0058
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## [1] "Levene's test for corolla_diam_mm :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
```

```
## group   1  0.7376 0.3913
##       241
## [1] "Mann-Whitney U test for corolla_diam_mm :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 4370, p-value = 4.422e-08
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for corolla_diam_mm :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -5.9403, df = 240.1, p-value = 9.929e-09
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -3.778892 -1.896752
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              22.74350              25.58132

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## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.56848, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.74156, p-value = 3.082e-14

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for skel_dryweight_mg_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  53.928 2.724e-12 ***
##       258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for skel_dryweight_mg_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
```

```
## data:  mean_trait by Year
## W = 1341.5, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for skel_dryweight_mg_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -9.6799, df = 141.99, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -200.3200 -132.3769
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              23.31637             189.66480

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## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.85469, p-value = 7.641e-09
##
## [1] "Shapiro-Wilk test for 2023 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.87028, p-value = 7.765e-09

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for SLA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.9659 0.3268
##       225
## [1] "Mann-Whitney U test for SLA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 5131, p-value = 0.009089
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for SLA_SEG :"
##
##  Welch Two Sample t-test
##
```

```
## data:  mean_trait by Year
## t = -1.8877, df = 214.27, p-value = 0.06042
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -2.78253325  0.06014569
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              16.75709              18.11828

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## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.9828, p-value = 0.133
##
## [1] "Shapiro-Wilk test for 2023 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.97714, p-value = 0.02031

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for d13C_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  2.5218 0.1135
##       255
## [1] "Mann-Whitney U test for d13C_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 6813, p-value = 0.01868
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for d13C_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -2.8574, df = 233.4, p-value = 0.004657
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -0.5843340 -0.1073889
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##             -29.71301             -29.36715
```

```
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## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.82306, p-value = 1.439e-08
##
## [1] "Shapiro-Wilk test for 2023 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.86648, p-value = 1.502e-09

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for msm_all :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  14.139 0.0002193 ***
##       213
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for msm_all :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 5696, p-value = 0.6243
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for msm_all :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = 1.7605, df = 111.07, p-value = 0.08107
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -0.06925685  1.17245871
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              4.126091              3.574490

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## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 LMA_SEG :"
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.93297, p-value = 4.152e-05
##
## [1] "Shapiro-Wilk test for 2023 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.74708, p-value = 4.294e-13

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for LMA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  1.8996 0.1695
##       225
## [1] "Mann-Whitney U test for LMA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 7591, p-value = 0.0178
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for LMA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = 1.7901, df = 223.62, p-value = 0.07479
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -0.0004840563  0.0100844294
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##            0.06603137            0.06123119

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## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.93601, p-value = 4.341e-06
##
## [1] "Shapiro-Wilk test for 2023 fl_duration :"
##
```

```
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.97732, p-value = 0.01778

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for fl_duration :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  21.129 6.468e-06 ***
##       284
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for fl_duration :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 1216.5, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for fl_duration :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -19.917, df = 244.5, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -38.24627 -31.36213
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              17.14219              51.94639

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## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.90819, p-value = 2.613e-06
##
## [1] "Shapiro-Wilk test for 2023 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.98, p-value = 0.0431

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
```

```
## factor.

## [1] "Levene's test for mean_seeds_per_fruit :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   1  3.0355 0.08276 .
##       237
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for mean_seeds_per_fruit :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 5251.5, p-value = 0.000924
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for mean_seeds_per_fruit :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -2.5961, df = 190.93, p-value = 0.01016
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -1.2461190 -0.1701026
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              2.628398              3.336508

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## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.36566, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.70997, p-value = 1.846e-15

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for est_fecundity :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group   1   49.22 1.69e-11 ***
```

```
##       283
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for est_fecundity :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 3062, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fecundity :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -7.998, df = 164.32, p-value = 2.145e-13
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -30.03489 -18.14134
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              3.775164             27.863279

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## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

## [1] "Shapiro-Wilk test for 2022 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.22584, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.51177, p-value < 2.2e-16

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for est_fitness :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  24.975 1.019e-06 ***
##       283
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for est_fitness :"
##
##  Wilcoxon rank sum test with continuity correction
```

```
##
## data:  mean_trait by Year
## W = 3394.5, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fitness :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -5.7395, df = 164.34, p-value = 4.47e-08
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -18.247291  -8.905967
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              2.135155             15.711784
```

#PATERNAL#2022v2023#

```r
# Define the traits to analyze
traits <- c("corolla_diam_mm", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG", "fl

# Loop over eBBh trait
for (trait in traits) {

  # Calculate the mean trait value for eBBh paternal (Donor) family for 2022
  paternal_means_22 <- BB_common_22_23_G1_22 %>%
    group_by(Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE))

  # Calculate the mean trait value for eBBh paternal (Donor) family for 2023 G1
  paternal_means_23_G1 <- BB_common_22_23_G1_23 %>%
    group_by(Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE))

  # Add year information
  paternal_means_22 <- paternal_means_22 %>%
    mutate(Year = "2022")

  paternal_means_23_G1 <- paternal_means_23_G1 %>%
    mutate(Year = "2023_G1")

  # Combine the dataframes
  pat_combined_data_year <- bind_rows(paternal_means_22, paternal_means_23_G1)

  # Create the histogram
  ggplot(pat_combined_data_year, aes(x = mean_trait, fill = Year)) +
    geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
    labs(title = paste("Comparison of Recipient Mean", trait),
         x = paste("Mean", trait),
         y = "Frequency") +
    scale_fill_manual(values = c("2022" = "blue", "2023_G1" = "red")) +
    theme_minimal()
```

```r
  # Split the data by year
  trait_2022 <- na.omit(paternal_means_22$mean_trait)
  trait_2023 <- na.omit(paternal_means_23_G1$mean_trait)

  # Check normality with Shapiro-Wilk test
  shapiro_test_2022 <- shapiro.test(trait_2022)
  shapiro_test_2023 <- shapiro.test(trait_2023)

  # Print Shapiro-Wilk test results
  print(paste("Shapiro-Wilk test for 2022", trait, ":"))
  print(shapiro_test_2022)

  print(paste("Shapiro-Wilk test for 2023", trait, ":"))
  print(shapiro_test_2023)

  # Perform Levene's test for homogeneity of variances
  levene_test <- leveneTest(mean_trait ~ Year, data = pat_combined_data_year)
  print(paste("Levene's test for", trait, ":"))
  print(levene_test)

  # Perform Mann-Whitney U test (Wilcoxon rank-sum test)
  mann_whitney_test <- wilcox.test(mean_trait ~ Year, data = pat_combined_data_year)
  print(paste("Mann-Whitney U test for", trait, ":"))
  print(mann_whitney_test)

  # Optionally perform t-test if data is normally distributed
 # if (shapiro_test_2022$p.value > 0.05 && shapiro_test_2023$p.value > 0.05 && levene_test$`Pr(>F)`[1] 
    t_test_result_year <- t.test(mean_trait ~ Year, data = pat_combined_data_year)
    print(paste("t-test for", trait, ":"))
    print(t_test_result_year)
  #} else {
  #  print(paste("Skipping t-test for", trait, "due to non-normal distribution or unequal variances"))
  }
```

```
## [1] "Shapiro-Wilk test for 2022 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.97904, p-value = 0.5256
##
## [1] "Shapiro-Wilk test for 2023 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.98015, p-value = 0.5717

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for corolla_diam_mm :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.1486 0.7007
```

```
##       96
## [1] "Mann-Whitney U test for corolla_diam_mm :"
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Year
## W = 509, p-value = 3.466e-07
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for corolla_diam_mm :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -5.7481, df = 94.866, p-value = 1.092e-07
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -3.627407 -1.764969
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              22.90077              25.59696
##
## [1] "Shapiro-Wilk test for 2022 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.80489, p-value = 1.382e-06
##
## [1] "Shapiro-Wilk test for 2023 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.83195, p-value = 6.224e-06

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for skel_dryweight_mg_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value    Pr(>F)
## group  1  26.191 1.586e-06 ***
##       96
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for skel_dryweight_mg_SEG :"
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Year
## W = 24, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for skel_dryweight_mg_SEG :"
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -9.5542, df = 50.41, p-value = 6.711e-13
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -190.5285 -124.3462
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              23.96312             181.40046
##
## [1] "Shapiro-Wilk test for 2022 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.95602, p-value = 0.06973
##
## [1] "Shapiro-Wilk test for 2023 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.96986, p-value = 0.239

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for SLA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  1.3768 0.2436
##       95
## [1] "Mann-Whitney U test for SLA_SEG :"
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Year
## W = 920, p-value = 0.06508
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for SLA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -1.6624, df = 89.283, p-value = 0.09994
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -2.3795112  0.2115752
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              16.68123              17.76520
##
```

```
## [1] "Shapiro-Wilk test for 2022 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.97761, p-value = 0.4696
##
## [1] "Shapiro-Wilk test for 2023 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.98651, p-value = 0.8422

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for d13C_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  3.0269 0.0851 .
##       96
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

## [1] "Mann-Whitney U test for d13C_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 858, p-value = 0.0151
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for d13C_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -2.7583, df = 93.16, p-value = 0.006993
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -0.54987287 -0.08954517
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##             -29.68409             -29.36438
##
## [1] "Shapiro-Wilk test for 2022 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.82518, p-value = 6.117e-06
##
```

```
## [1] "Shapiro-Wilk test for 2023 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.90765, p-value = 0.0009864

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for msm_all :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value    Pr(>F)
## group  1  12.342 0.0006827 ***
##       94
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for msm_all :"
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Year
## W = 1163, p-value = 0.936
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for msm_all :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = 1.6771, df = 59.111, p-value = 0.0988
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -0.1105322  1.2555751
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              4.136096              3.563574
##
## [1] "Shapiro-Wilk test for 2022 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.91586, p-value = 0.002131
##
## [1] "Shapiro-Wilk test for 2023 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.64961, p-value = 1.471e-09

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for LMA_SEG :"
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.1924 0.6619
##       95
## [1] "Mann-Whitney U test for LMA_SEG :"
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Year
## W = 1449, p-value = 0.04897
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for LMA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = 0.90201, df = 93.153, p-value = 0.3694
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal t
## 95 percent confidence interval:
##  -0.003354456  0.008938338
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##            0.06599094            0.06319900
##
## [1] "Shapiro-Wilk test for 2022 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.92257, p-value = 0.003273
##
## [1] "Shapiro-Wilk test for 2023 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.97406, p-value = 0.3485

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for fl_duration :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value  Pr(>F)
## group  1   5.391 0.02236 *
##       96
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

## [1] "Mann-Whitney U test for fl_duration :"
##
##  Wilcoxon rank sum test with continuity correction
```

```
##
## data:  mean_trait by Year
## W = 10, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for fl_duration :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -19.533, df = 84.137, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -38.78873 -31.62063
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              16.99886              52.20354
##
## [1] "Shapiro-Wilk test for 2022 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.92135, p-value = 0.003318
##
## [1] "Shapiro-Wilk test for 2023 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.98395, p-value = 0.7366

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for mean_seeds_per_fruit :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value  Pr(>F)
## group  1  3.9054 0.05103 .
##       95
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

## [1] "Mann-Whitney U test for mean_seeds_per_fruit :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 755.5, p-value = 0.002439
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for mean_seeds_per_fruit :"
##
```

```
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -2.3275, df = 82.531, p-value = 0.02239
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -1.23575801 -0.09687839
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              2.652389              3.318707
##
## [1] "Shapiro-Wilk test for 2022 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.47669, p-value = 5.636e-12
##
## [1] "Shapiro-Wilk test for 2023 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.8533, p-value = 2.249e-05

## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## [1] "Levene's test for est_fecundity :"
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value    Pr(>F)
## group  1  22.946 6.057e-06 ***
##       96
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties

## [1] "Mann-Whitney U test for est_fecundity :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Year
## W = 105, p-value = 7.228e-15
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fecundity :"
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Year
## t = -7.7653, df = 58.791, p-value = 1.396e-10
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -28.90086 -17.05731
```

```
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              3.936806             26.915894
```

```
# }
```

#CALCULATING PATERNAL FITNESS ESTIMATES

```
# Define the datasets and the years they correspond to
datasets <- list(
  "2022" = BB_common_22_23_G1_22,
  "2023_G1" = BB_common_22_23_G1_23
)

# Initialize an empty list to store results
paternal_means_list <- list()

# Loop over each dataset and perform the calculations
for (year in names(datasets)) {
  # Calculate the mean trait value for each maternal (Recipient) family
  maternal_means <- datasets[[year]] %>%
    group_by(Recipient, Donor) %>%
    summarise(mat_mean_est_fitness = mean(est_fitness, na.rm = TRUE))

  # Calculate the mean est_fitness for each Donor
  paternal_means <- maternal_means %>%
    group_by(Donor) %>%
    summarise(pat_mean_est_fitness = mean(mat_mean_est_fitness, na.rm = TRUE)) %>%
    mutate(Year = year)

  # Store the result in the list
  paternal_means_list[[year]] <- paternal_means
}
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
# Combine the dataframes for different years
pat_combined_data_year <- bind_rows(paternal_means_list)

# Split the data by year
trait_2022 <- na.omit(paternal_means_list[["2022"]]$pat_mean_est_fitness)
trait_2023 <- na.omit(paternal_means_list[["2023_G1"]]$pat_mean_est_fitness)

# Check normality with Shapiro-Wilk test
shapiro_test_2022 <- shapiro.test(trait_2022)
shapiro_test_2023 <- shapiro.test(trait_2023)

# Print Shapiro-Wilk test results
print("Shapiro-Wilk test for 2022:")
```

```
## [1] "Shapiro-Wilk test for 2022:"
```

```
print(shapiro_test_2022)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.38866, p-value = 5.38e-13
```

```r
print("Shapiro-Wilk test for 2023:")
```

```
## [1] "Shapiro-Wilk test for 2023:"
```

```r
print(shapiro_test_2023)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.67261, p-value = 3.516e-09
```

```r
# Perform Levene's test for homogeneity of variances
levene_test <- leveneTest(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```r
print("Levene's test:")
```

```
## [1] "Levene's test:"
```

```r
print(levene_test)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value    Pr(>F)
## group  1  12.445 0.0006449 ***
##       96
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Perform Mann-Whitney U test (Wilcoxon rank-sum test)
mann_whitney_test <- wilcox.test(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
```

```r
print("Mann-Whitney U test:")
```

```
## [1] "Mann-Whitney U test:"
```

```r
print(mann_whitney_test)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  pat_mean_est_fitness by Year
## W = 133, p-value = 3.42e-14
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Optionally perform t-test if data is normally distributed
# if (shapiro_test_2022$p.value > 0.05 && shapiro_test_2023$p.value > 0.05 && levene_test$`Pr(>F)`[1] >
t_test_result_year <- t.test(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
print("t-test:")
```

```
## [1] "t-test:"
```

```r
print(t_test_result_year)
```

```
##
##  Welch Two Sample t-test
##
## data:  pat_mean_est_fitness by Year
## t = -5.4451, df = 54.319, p-value = 1.282e-06
## alternative hypothesis: true difference in means between group 2022 and group 2023_G1 is not equal to
## 95 percent confidence interval:
##  -18.222465  -8.415651
## sample estimates:
##    mean in group 2022 mean in group 2023_G1
##              2.130089             15.449147
```

```r
# } else {
#   print("Skipping t-test due to non-normal distribution or unequal variances")
# }
```

#PHENOTYPE#2022v2023

```r
# Define the traits to analyze
traits <- c("corolla_diam_mm", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG", "fl

# Loop over eBBh trait
for (trait in traits) {

  # Convert 'Year' to factor if it's not already
  if (!is.factor(BB_common_22_23_G1$Year)) {
    BB_common_22_23_G1$Year <- as.factor(BB_common_22_23_G1$Year)
  }

  # Create the histogram
  ggplot(BB_common_22_23_G1, aes_string(x = trait, fill = "Year")) +
    geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
    labs(title = paste("Comparison of", trait),
         x = trait,
         y = "Frequency") +
    scale_fill_manual(values = c("2022" = "blue", "2023" = "red")) +
    theme_minimal()

  # Split the data by year
  trait_2022 <- na.omit(BB_common_22_23_G1_22[[trait]])
  trait_2023 <- na.omit(BB_common_22_23_G1_23[[trait]])

  # Check normality with Shapiro-Wilk test
  shapiro_test_2022 <- shapiro.test(trait_2022)
  shapiro_test_2023 <- shapiro.test(trait_2023)

  # Print Shapiro-Wilk test results
  print(paste("Shapiro-Wilk test for 2022", trait, ":"))
  print(shapiro_test_2022)

  print(paste("Shapiro-Wilk test for 2023", trait, ":"))
  print(shapiro_test_2023)
```

```r
# Perform Levene's test for homogeneity of variances
levene_test <- leveneTest(as.formula(paste(trait, "~ as.factor(Year)")), data = BB_common_22_23_G1)
print(paste("Levene's test for", trait, ":"))
print(levene_test)

# Perform Mann-Whitney U test (Wilcoxon rank-sum test)
mann_whitney_test <- wilcox.test(as.formula(paste(trait, "~ Year")), data = BB_common_22_23_G1)
print(paste("Mann-Whitney U test for", trait, ":"))
print(mann_whitney_test)

# Optionally perform t-test if data is normally distributed and variances are equal
#if (shapiro_test_2022$p.value > 0.05 && shapiro_test_2023$p.value > 0.05 && levene_test$`Pr(>F)`[1] 
  t_test_result_year <- t.test(as.formula(paste(trait, "~ Year")), data = BB_common_22_23_G1)
  print(paste("t-test for", trait, ":"))
  print(t_test_result_year)
} #else {
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## [1] "Shapiro-Wilk test for 2022 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.99506, p-value = 0.7049
##
## [1] "Shapiro-Wilk test for 2023 corolla_diam_mm :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.99006, p-value = 0.1965
##
## [1] "Levene's test for corolla_diam_mm :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group   1  7.6599 0.005902 **
##       410
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for corolla_diam_mm :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  corolla_diam_mm by Year
## W = 14687, p-value = 8.261e-08
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for corolla_diam_mm :"
```

```
##
##  Welch Two Sample t-test
##
## data:  corolla_diam_mm by Year
## t = -5.7334, df = 408.56, p-value = 1.916e-08
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -3.35387 -1.64122
## sample estimates:
## mean in group 2022 mean in group 2023
##           22.93347           25.43102
##
## [1] "Shapiro-Wilk test for 2022 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.47189, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.57256, p-value < 2.2e-16
##
## [1] "Levene's test for skel_dryweight_mg_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  46.299 3.123e-11 ***
##       467
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for skel_dryweight_mg_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  skel_dryweight_mg_SEG by Year
## W = 6447, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for skel_dryweight_mg_SEG :"
##
##  Welch Two Sample t-test
##
## data:  skel_dryweight_mg_SEG by Year
## t = -9.5464, df = 286.51, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -193.2888 -127.2084
## sample estimates:
## mean in group 2022 mean in group 2023
##           24.01423          184.26286
##
```

```
## [1] "Shapiro-Wilk test for 2022 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.87925, p-value = 3.933e-09
##
## [1] "Shapiro-Wilk test for 2023 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.89781, p-value = 2.658e-10
##
## [1] "Levene's test for SLA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2115 0.6459
##       329
## [1] "Mann-Whitney U test for SLA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  SLA_SEG by Year
## W = 11216, p-value = 0.01705
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for SLA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  SLA_SEG by Year
## t = -1.6089, df = 285, p-value = 0.1088
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -2.2771980  0.2288267
## sample estimates:
## mean in group 2022 mean in group 2023
##           16.84627           17.87046
##
## [1] "Shapiro-Wilk test for 2022 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.99085, p-value = 0.2402
##
## [1] "Shapiro-Wilk test for 2023 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.99315, p-value = 0.2199
##
```

```
## [1] "Levene's test for d13C_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1   2.132 0.1449
##       481
## [1] "Mann-Whitney U test for d13C_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  d13C_SEG by Year
## W = 24976, p-value = 0.02972
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for d13C_SEG :"
##
##  Welch Two Sample t-test
##
## data:  d13C_SEG by Year
## t = -2.3497, df = 394.09, p-value = 0.01928
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -0.45998799 -0.04089318
## sample estimates:
## mean in group 2022 mean in group 2023
##          -29.62206          -29.37162
##
## [1] "Shapiro-Wilk test for 2022 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.82127, p-value = 4.131e-10
##
## [1] "Shapiro-Wilk test for 2023 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.87126, p-value = 3.197e-13
##
## [1] "Levene's test for msm_all :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group   1  7.5337 0.006375 **
##       342
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for msm_all :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  msm_all by Year
## W = 12888, p-value = 0.8673
## alternative hypothesis: true location shift is not equal to 0
```

```
##
## [1] "t-test for msm_all :"
##
##  Welch Two Sample t-test
##
## data:  msm_all by Year
## t = 1.3155, df = 152.7, p-value = 0.1903
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -0.1706496  0.8508100
## sample estimates:
## mean in group 2022 mean in group 2023
##            3.92645            3.58637
##
## [1] "Shapiro-Wilk test for 2022 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.9506, p-value = 8.651e-05
##
## [1] "Shapiro-Wilk test for 2023 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.48297, p-value < 2.2e-16
##
## [1] "Levene's test for LMA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1   0.044  0.834
##       329
## [1] "Mann-Whitney U test for LMA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  LMA_SEG by Year
## W = 15304, p-value = 0.01705
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for LMA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  LMA_SEG by Year
## t = 0.94979, df = 324.9, p-value = 0.3429
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -0.002944477  0.008441482
## sample estimates:
## mean in group 2022 mean in group 2023
##         0.06534875         0.06260025
##
```

```
## [1] "Shapiro-Wilk test for 2022 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.88766, p-value = 3.171e-15
##
## [1] "Shapiro-Wilk test for 2023 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.95473, p-value = 8.723e-09
##
## [1] "Levene's test for fl_duration :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  39.084 7.133e-10 ***
##       686
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for fl_duration :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  fl_duration by Year
## W = 13663, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for fl_duration :"
##
##  Welch Two Sample t-test
##
## data:  fl_duration by Year
## t = -23.408, df = 609.22, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -38.29171 -32.36397
## sample estimates:
## mean in group 2022 mean in group 2023
##           16.72464           52.05248
##
## [1] "Shapiro-Wilk test for 2022 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.84329, p-value = 2.573e-11
##
## [1] "Shapiro-Wilk test for 2023 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
```

```
## W = 0.96658, p-value = 9.31e-06
##
## [1] "Levene's test for mean_seeds_per_fruit :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  1.1574 0.2826
##       407
## [1] "Mann-Whitney U test for mean_seeds_per_fruit :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_seeds_per_fruit by Year
## W = 15154, p-value = 0.0002398
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for mean_seeds_per_fruit :"
##
##  Welch Two Sample t-test
##
## data:  mean_seeds_per_fruit by Year
## t = -2.3543, df = 257.19, p-value = 0.01931
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -1.1656677 -0.1038302
## sample estimates:
## mean in group 2022 mean in group 2023
##           2.708744           3.343493
##
## [1] "Shapiro-Wilk test for 2022 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.1871, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.59303, p-value < 2.2e-16
##
## [1] "Levene's test for est_fecundity :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  67.487 1.084e-15 ***
##       674
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for est_fecundity :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  est_fecundity by Year
```

```
## W = 28975, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fecundity :"
##
##  Welch Two Sample t-test
##
## data:  est_fecundity by Year
## t = -8.4063, df = 427.39, p-value = 6.363e-16
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -28.94674 -17.97551
## sample estimates:
## mean in group 2022 mean in group 2023
##           3.887978          27.349106
##
## [1] "Shapiro-Wilk test for 2022 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2022
## W = 0.11428, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for 2023 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023
## W = 0.41994, p-value < 2.2e-16
##
## [1] "Levene's test for est_fitness :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  35.904 3.375e-09 ***
##       674
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## [1] "Mann-Whitney U test for est_fitness :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  est_fitness by Year
## W = 29736, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fitness :"
##
##  Welch Two Sample t-test
##
## data:  est_fitness by Year
## t = -6.1592, df = 448.19, p-value = 1.627e-09
## alternative hypothesis: true difference in means between group 2022 and group 2023 is not equal to 0
## 95 percent confidence interval:
##  -16.868869  -8.707918
```

```
## sample estimates:
## mean in group 2022 mean in group 2023
##           2.282322          15.070715
```

```
    #print(paste("Skipping t-test for", trait, "due to non-normal distribution or unequal variances"))
  #}
#}
```

Determine if G1 and G2 trait means are significantly different in 2023 Any difference would be due to either the effect of greenhouse produced seeds or a generation of selection

#MATERNAL#G1vG2#

```
# Define the traits to analyze
traits <- c("corolla_diam_mm_SEG", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG"

# Loop over eBBh trait
for (trait in traits) {

  # Create a dataframe with common "Genotype" values for 2023
  BB_common_23_G1 <- semi_join(BB_23_fit_G1, BB_23_fit_G2, by = "Genotype")
  BB_common_23_G2 <- semi_join(BB_23_fit_G2, BB_23_fit_G1, by = "Genotype")

  # Merge these two dataframes to have one finalized dataset to compare G1 2023 and G2 2023
  BB_common_23_G1_G2 <- full_join(BB_common_23_G1, BB_common_23_G2, by = intersect(names(BB_common_23_G

  # Calculate the mean trait value for eBBh maternal (Recipient) family for G1, 2023
  maternal_means_23_G1 <- BB_common_23_G1 %>%
    group_by(Recipient, Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE)) %>%
    mutate(Gen = "1")

  # Calculate the mean trait value for eBBh maternal (Recipient) family for G2, 2023
  maternal_means_23_G2 <- BB_common_23_G2 %>%
    group_by(Recipient, Donor) %>%
    summarise(mean_trait = mean(get(trait), na.rm = TRUE)) %>%
    mutate(Gen = "2")

  # Combine the dataframes
  mat_combined_data_gen <- bind_rows(maternal_means_23_G1, maternal_means_23_G2)

  # Create the histogram
  ggplot(mat_combined_data_gen, aes(x = mean_trait, fill = Gen)) +
    geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
    labs(title = paste("Comparison of Recipient Mean", trait, ": G1 vs G2"),
         x = paste("Mean", trait),
         y = "Frequency") +
    scale_fill_manual(values = c("1" = "blue", "2" = "red")) +
    theme_minimal()

  # Split the data by generation
  trait_G1 <- na.omit(maternal_means_23_G1$mean_trait)
  trait_G2 <- na.omit(maternal_means_23_G2$mean_trait)

  # Check normality with Shapiro-Wilk test
  shapiro_test_G1 <- shapiro.test(trait_G1)
```

```r
  shapiro_test_G2 <- shapiro.test(trait_G2)

  # Print Shapiro-Wilk test results
  print(shapiro_test_G1)
  print(shapiro_test_G2)

  # Perform Levene's test for homogeneity of variances
  levene_test <- leveneTest(mean_trait ~ as.factor(Gen), data = mat_combined_data_gen)

  # Print Levene's test result
  print(levene_test)

  # Perform Mann-Whitney U test (Wilcoxon rank-sum test) instead of t-test
  mann_whitney_test <- wilcox.test(mean_trait ~ Gen, data = mat_combined_data_gen)

  # Print Mann-Whitney test result
  print(mann_whitney_test)

  # Perform t-test
  t_test_result_gen <- t.test(mean_trait ~ Gen, data = mat_combined_data_gen)

  # Print t-test result
  print(t_test_result_gen)

}
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.96194, p-value = 0.0268
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.98934, p-value = 0.8139
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.5867  0.445
##       142
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 1936, p-value = 0.008862
## alternative hypothesis: true location shift is not equal to 0
##
```

```
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -2.6878, df = 141.58, p-value = 0.008055
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -2.5279231 -0.3852767
## sample estimates:
## mean in group 1 mean in group 2
##        24.84538        26.30198
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.73148, p-value = 2.52e-10
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.72275, p-value = 1.339e-10
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0047 0.9457
##       147
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 2985, p-value = 0.4264
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.43886, df = 146.76, p-value = 0.6614
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -51.54963  80.97949
## sample estimates:
## mean in group 1 mean in group 2
##        191.3831        176.6681
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
```

```
## `.groups` argument.
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.80436, p-value = 1.025e-07
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.88775, p-value = 4.844e-05
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2835 0.5954
##       121
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 1887, p-value = 0.9899
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.42321, df = 117.62, p-value = 0.6729
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -1.503737  2.321145
## sample estimates:
## mean in group 1 mean in group 2
##        18.15038        17.74168
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.9633, p-value = 0.02862
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.99337, p-value = 0.9667
##
## Levene's Test for Homogeneity of Variance (center = median)
```

```
##        Df F value Pr(>F)
## group   1  0.3475 0.5564
##       148
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 2780, p-value = 0.9043
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.33168, df = 146.49, p-value = 0.7406
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.2513406  0.3527184
## sample estimates:
## mean in group 1 mean in group 2
##       -29.34109       -29.39178
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.78799, p-value = 9.792e-09
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.8118, p-value = 6.927e-08
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  1.1531 0.2848
##       137
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 2065, p-value = 0.142
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -1.2911, df = 129.94, p-value = 0.199
```

```
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.9225477  0.1939285
## sample estimates:
## mean in group 1 mean in group 2
##        3.545502        3.909812

## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.85135, p-value = 2.101e-06
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.98741, p-value = 0.7933
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.2908 0.5907
##       121
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 1897, p-value = 0.9738
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.25265, df = 113.49, p-value = 0.801
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.005332251  0.006891117
## sample estimates:
## mean in group 1 mean in group 2
##      0.06142522      0.06064579

## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.

##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
```

```
## W = 0.94255, p-value = 0.001687
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.94708, p-value = 0.002963
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.5302 0.4676
##       152
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 2901.5, p-value = 0.8211
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -0.053085, df = 151.47, p-value = 0.9577
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -5.459773  5.174059
## sample estimates:
## mean in group 1 mean in group 2
##        49.84416        49.98701
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.98095, p-value = 0.328
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.96247, p-value = 0.0305
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0753 0.7842
##       144
##
##  Wilcoxon rank sum test with continuity correction
##
```

```
## data:  mean_trait by Gen
## W = 2659.5, p-value = 0.9875
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -0.19461, df = 142.28, p-value = 0.846
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.7306338  0.5996669
## sample estimates:
## mean in group 1 mean in group 2
##        3.602173        3.667657
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.87901, p-value = 2.56e-06
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.66137, p-value = 5.942e-12
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.9723 0.3257
##       151
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 3044.5, p-value = 0.6666
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -0.75743, df = 121.2, p-value = 0.4503
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -17.897916   7.992434
## sample estimates:
## mean in group 1 mean in group 2
##        30.16229        35.11503
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.82973, p-value = 5.58e-08
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.63325, p-value = 1.752e-12
##
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.4794 0.4898
##       151
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 2994, p-value = 0.8054
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -0.64521, df = 124.44, p-value = 0.52
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -10.029175   5.097889
## sample estimates:
## mean in group 1 mean in group 2
##        15.78650        18.25214
```

#PATERNAL#G1vG2#

```
# Define the traits to analyze
traits <- c("corolla_diam_mm_SEG", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG"

# Loop over eBBh trait
for (trait in traits) {

  # Create a dataframe with common "Genotype" values for 2023
  BB_common_23_G1 <- semi_join(BB_23_fit_G1, BB_23_fit_G2, by = "Genotype")
  BB_common_23_G2 <- semi_join(BB_23_fit_G2, BB_23_fit_G1, by = "Genotype")

  # Merge these two dataframes to have one finalized dataset to compare G1 2023 and G2 2023
  BB_common_23_G1_G2 <- full_join(BB_common_23_G1, BB_common_23_G2, by = intersect(names(BB_common_23_G
```

```r
# Calculate the mean trait value for eBBh paternal (Donor) family for G1, 2023
paternal_means_23_G1 <- BB_common_23_G1 %>%
  group_by(Donor) %>%
  summarise(mean_trait = mean(get(trait), na.rm = TRUE), .groups = "drop") %>%
  mutate(Gen = "1")

# Calculate the mean trait value for eBBh paternal (Donor) family for G2, 2023
paternal_means_23_G2 <- BB_common_23_G2 %>%
  group_by(Donor) %>%
  summarise(mean_trait = mean(get(trait), na.rm = TRUE), .groups = "drop") %>%
  mutate(Gen = "2")

# Combine the dataframes
pat_combined_data_gen <- bind_rows(paternal_means_23_G1, paternal_means_23_G2)

# Create the histogram
ggplot(pat_combined_data_gen, aes(x = mean_trait, fill = Gen)) +
  geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
  labs(title = paste("Comparison of Donor Mean", trait, ": G1 vs G2"),
       x = paste("Mean", trait),
       y = "Frequency") +
  scale_fill_manual(values = c("1" = "blue", "2" = "red")) +
  theme_minimal()

# Split the data by generation and ensure numeric conversion
trait_G1 <- na.omit(as.numeric(paternal_means_23_G1$mean_trait))
trait_G2 <- na.omit(as.numeric(paternal_means_23_G2$mean_trait))

# Check normality with Shapiro-Wilk test if sample size is appropriate
if (length(trait_G1) >= 3 && length(trait_G1) <= 5000) {
  shapiro_test_G1 <- shapiro.test(trait_G1)
  print(shapiro_test_G1)
} else {
  print(paste("Sample size for trait_G1 is not suitable for Shapiro-Wilk test:", length(trait_G1)))
}

if (length(trait_G2) >= 3 && length(trait_G2) <= 5000) {
  shapiro_test_G2 <- shapiro.test(trait_G2)
  print(shapiro_test_G2)
} else {
  print(paste("Sample size for trait_G2 is not suitable for Shapiro-Wilk test:", length(trait_G2)))
}

# Perform Levene's test for homogeneity of variances
levene_test <- leveneTest(mean_trait ~ as.factor(Gen), data = pat_combined_data_gen)
print(levene_test)

# Perform Mann-Whitney U test (Wilcoxon rank-sum test)
mann_whitney_test <- wilcox.test(mean_trait ~ Gen, data = pat_combined_data_gen)
print(mann_whitney_test)

# Perform t-test
t_test_result_gen <- t.test(mean_trait ~ Gen, data = pat_combined_data_gen)
```

```r
  print(t_test_result_gen)
}
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.97634, p-value = 0.4942
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.95915, p-value = 0.129
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.1202 0.7297
##       85
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Gen
## W = 632, p-value = 0.007335
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -2.5529, df = 84.489, p-value = 0.01248
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -2.2783950 -0.2831879
## sample estimates:
## mean in group 1 mean in group 2
##        25.04957        26.33036
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.91319, p-value = 0.00284
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.86675, p-value = 0.00012
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.0056 0.9407
##       86
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Gen
## W = 1086, p-value = 0.3285
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.63781, df = 86, p-value = 0.5253
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -36.06124  70.13271
## sample estimates:
## mean in group 1 mean in group 2
##        179.7447        162.7090
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.95126, p-value = 0.08382
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.80049, p-value = 6.935e-06
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.1281 0.7214
##       78
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Gen
## W = 800, p-value = 1
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -0.35763, df = 72.185, p-value = 0.7217
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -2.241305  1.559419
## sample estimates:
## mean in group 1 mean in group 2
##        17.79513        18.13608
##
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.98144, p-value = 0.6913
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.96851, p-value = 0.2678
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.5604 0.4561
##       86
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 1024.5, p-value = 0.6403
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.46452, df = 85.748, p-value = 0.6435
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.2262020  0.3641409
## sample estimates:
## mean in group 1 mean in group 2
##       -29.35972       -29.42869
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.77827, p-value = 1.015e-06
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.92651, p-value = 0.009896
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.5556 0.4581
##       84
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Gen
## W = 731, p-value = 0.09644
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = -1.1369, df = 83.791, p-value = 0.2588
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.8825899  0.2405269
## sample estimates:
## mean in group 1 mean in group 2
##        3.569858        3.890889
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.79656, p-value = 5.747e-06
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.97273, p-value = 0.4371
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1   2e-04 0.9887
##       78
##
##  Wilcoxon rank sum exact test
##
## data:  mean_trait by Gen
## W = 824, p-value = 0.8221
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.82188, df = 75.248, p-value = 0.4137
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.003676974  0.008842344
## sample estimates:
## mean in group 1 mean in group 2
##      0.06218265      0.05959996
##
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.97483, p-value = 0.4423
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.98547, p-value = 0.8455
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.0517 0.8207
##       86
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 1037.5, p-value = 0.5647
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.48737, df = 85.984, p-value = 0.6272
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -3.857370  6.363051
## sample estimates:
## mean in group 1 mean in group 2
##        51.13774        49.88490
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.97984, p-value = 0.6273
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.95084, p-value = 0.06921
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  1.0561  0.307
##       84
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 948.5, p-value = 0.8357
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.5771, df = 82.263, p-value = 0.5654
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
## 95 percent confidence interval:
##  -0.3990699  0.7252485
## sample estimates:
## mean in group 1 mean in group 2
##        3.627989        3.464899
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.89259, p-value = 0.0006493
##
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.88865, p-value = 0.0005764
##
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.0023 0.9621
##       85
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_trait by Gen
## W = 964, p-value = 0.8819
## alternative hypothesis: true location shift is not equal to 0
##
##
##  Welch Two Sample t-test
##
## data:  mean_trait by Gen
## t = 0.0060227, df = 84.768, p-value = 0.9952
## alternative hypothesis: true difference in means between group 1 and group 2 is not equal to 0
```

```
## 95 percent confidence interval:
##  -9.98606 10.04674
## sample estimates:
## mean in group 1 mean in group 2
##        29.99741        29.96707
```

#CALCULATING PATERNAL FITNESS ESTIMATES

```r
# Define the datasets and the years they correspond to
datasets <- list(
  "2023_G1" = BB_common_23_G1,
  "2023_G2" = BB_common_23_G2
)

# Initialize an empty list to store results
paternal_means_list <- list()

# Loop over each dataset and perform the calculations
for (year in names(datasets)) {
  # Calculate the mean trait value for each maternal (Recipient) family
  maternal_means <- datasets[[year]] %>%
    group_by(Recipient, Donor) %>%
    summarise(mat_mean_est_fitness = mean(est_fitness, na.rm = TRUE))

  # Calculate the mean est_fitness for each Donor
  paternal_means <- maternal_means %>%
    group_by(Donor) %>%
    summarise(pat_mean_est_fitness = mean(mat_mean_est_fitness, na.rm = TRUE)) %>%
    mutate(Year = year)

  # Store the result in the list
  paternal_means_list[[year]] <- paternal_means
}
```

```
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
## `summarise()` has grouped output by 'Recipient'. You can override using the
## `.groups` argument.
```

```r
# Combine the dataframes for different years
pat_combined_data_year <- bind_rows(paternal_means_list)

# Split the data by year
trait_2023_G1 <- na.omit(paternal_means_list[["2023_G1"]]$pat_mean_est_fitness)
trait_2023_G2 <- na.omit(paternal_means_list[["2023_G2"]]$pat_mean_est_fitness)

# Check normality with Shapiro-Wilk test
shapiro_test_2023_G1 <- shapiro.test(trait_2023_G1)
shapiro_test_2023_G2 <- shapiro.test(trait_2023_G2)

# Print Shapiro-Wilk test results
print("Shapiro-Wilk test for 2023 G1:")
```

```
## [1] "Shapiro-Wilk test for 2023 G1:"
```

```r
print(shapiro_test_2023_G1)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023_G1
## W = 0.86517, p-value = 0.0001088
```

```r
print("Shapiro-Wilk test for 2023 G2:")
```

```
## [1] "Shapiro-Wilk test for 2023 G2:"
```

```r
print(shapiro_test_2023_G2)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  trait_2023_G2
## W = 0.79587, p-value = 2.931e-06
```

```r
# Perform Levene's test for homogeneity of variances
levene_test <- leveneTest(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```r
print("Levene's test:")
```

```
## [1] "Levene's test:"
```

```r
print(levene_test)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  1  0.6864 0.4097
##       85
```

```r
# Perform Mann-Whitney U test (Wilcoxon rank-sum test)
mann_whitney_test <- wilcox.test(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
## compute exact p-value with ties
```

```r
print("Mann-Whitney U test:")
```

```
## [1] "Mann-Whitney U test:"
```

```r
print(mann_whitney_test)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  pat_mean_est_fitness by Year
## W = 978.5, p-value = 0.7859
## alternative hypothesis: true location shift is not equal to 0
```

```r
# Optionally perform t-test if data is normally distributed
# if (shapiro_test_2022$p.value > 0.05 && shapiro_test_2023$p.value > 0.05 && levene_test$`Pr(>F)`[1] >
t_test_result_year <- t.test(pat_mean_est_fitness ~ Year, data = pat_combined_data_year)
print("t-test:")
```

```
## [1] "t-test:"
```

```r
print(t_test_result_year)
```

```
##
##  Welch Two Sample t-test
##
## data:  pat_mean_est_fitness by Year
## t = -0.36569, df = 76.955, p-value = 0.7156
## alternative hypothesis: true difference in means between group 2023_G1 and group 2023_G2 is not equal
## 95 percent confidence interval:
##  -8.152901  5.622989
## sample estimates:
## mean in group 2023_G1 mean in group 2023_G2
##              15.44440              16.70936
```

```r
# } else {
#   print("Skipping t-test due to non-normal distribution or unequal variances")
# }
```

#PHENOTYPE#G1vG2

```r
# Define the traits to analyze
traits <- c("corolla_diam_mm_SEG", "skel_dryweight_mg_SEG", "SLA_SEG", "d13C_SEG", "msm_all", "LMA_SEG"

# Loop over eBBh trait
for (trait in traits) {

  # Create a dataframe with common "Genotype" values for 2023
  BB_common_23_G1 <- semi_join(BB_23_fit_G1, BB_23_fit_G2, by = "Genotype")
  BB_common_23_G2 <- semi_join(BB_23_fit_G2, BB_23_fit_G1, by = "Genotype")

  # Merge these two dataframes to have one finalized dataset to compare G1 2023 and G2 2023
  BB_common_23_G1_G2 <- full_join(BB_common_23_G1, BB_common_23_G2, by = intersect(names(BB_common_23_G

  # Convert 'Gen' to factor if it's not already
  if (!is.factor(BB_common_23_G1_G2$Gen)) {
    BB_common_23_G1_G2$Gen <- as.factor(BB_common_23_G1_G2$Gen)
  }

  # Create the histogram
  ggplot(BB_common_23_G1_G2, aes_string(x = trait, fill = "Gen")) +
    geom_histogram(position = "identity", alpha = 0.6, binwidth = 0.2) +
    labs(title = paste("Comparison of", trait),
         x = trait,
         y = "Frequency") +
    scale_fill_manual(values = c("1" = "blue", "2" = "red")) +
    theme_minimal()

  # Split the data by generation and ensure numeric conversion
  trait_G1 <- na.omit(as.numeric(BB_common_23_G1[[trait]]))
  trait_G2 <- na.omit(as.numeric(BB_common_23_G2[[trait]]))

  # Check normality with Shapiro-Wilk test if sample size is appropriate
  if (length(trait_G1) >= 3 && length(trait_G1) <= 5000) {
    shapiro_test_G1 <- shapiro.test(trait_G1)
```

```r
    print(paste("Shapiro-Wilk test for G1", trait, ":"))
    print(shapiro_test_G1)
  } else {
    print(paste("Sample size for G1", trait, "is not suitable for Shapiro-Wilk test:", length(trait_G1)
  }

  if (length(trait_G2) >= 3 && length(trait_G2) <= 5000) {
    shapiro_test_G2 <- shapiro.test(trait_G2)
    print(paste("Shapiro-Wilk test for G2", trait, ":"))
    print(shapiro_test_G2)
  } else {
    print(paste("Sample size for G2", trait, "is not suitable for Shapiro-Wilk test:", length(trait_G2)
  }

  # Perform Levene's test for homogeneity of variances
  levene_test <- leveneTest(as.formula(paste(trait, "~ as.factor(Gen)")), data = BB_common_23_G1_G2)
  print(paste("Levene's test for", trait, ":"))
  print(levene_test)

  # Perform Mann-Whitney U test (Wilcoxon rank-sum test)
  mann_whitney_test <- wilcox.test(as.formula(paste(trait, "~ Gen")), data = BB_common_23_G1_G2)
  print(paste("Mann-Whitney U test for", trait, ":"))
  print(mann_whitney_test)

  # Perform t-test
  t_test_result_gen <- t.test(as.formula(paste(trait, "~ Gen")), data = BB_common_23_G1_G2)
  print(paste("t-test for", trait, ":"))
  print(t_test_result_gen)
}
```

```
## [1] "Shapiro-Wilk test for G1 corolla_diam_mm_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.99554, p-value = 0.9228
##
## [1] "Shapiro-Wilk test for G2 corolla_diam_mm_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.98303, p-value = 0.1258
##
## [1] "Levene's test for corolla_diam_mm_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0036  0.952
##       276
## [1] "Mann-Whitney U test for corolla_diam_mm_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  corolla_diam_mm_SEG by Gen
```

```
## W = 7711, p-value = 0.006235
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for corolla_diam_mm_SEG :"
##
##  Welch Two Sample t-test
##
## data:  corolla_diam_mm_SEG by Gen
## t = -3.1133, df = 267.37, p-value = 0.002051
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -2.4591626 -0.5537717
## sample estimates:
## mean in group G1 mean in group G2
##         24.83226         26.33873
##
## [1] "Shapiro-Wilk test for G1 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.62905, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for G2 skel_dryweight_mg_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.67328, p-value = 1.478e-15
##
## [1] "Levene's test for skel_dryweight_mg_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0389 0.8438
##       274
## [1] "Mann-Whitney U test for skel_dryweight_mg_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  skel_dryweight_mg_SEG by Gen
## W = 9819.5, p-value = 0.61
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for skel_dryweight_mg_SEG :"
##
##  Welch Two Sample t-test
##
## data:  skel_dryweight_mg_SEG by Gen
## t = 0.46003, df = 273.98, p-value = 0.6459
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -43.95674  70.76458
## sample estimates:
## mean in group G1 mean in group G2
```

```
##         183.3114         169.9074
##
## [1] "Shapiro-Wilk test for G1 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.86505, p-value = 3.516e-08
##
## [1] "Shapiro-Wilk test for G2 SLA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.88273, p-value = 1.504e-06
##
## [1] "Levene's test for SLA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.1155 0.7343
##       184
## [1] "Mann-Whitney U test for SLA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  SLA_SEG by Gen
## W = 4314, p-value = 0.9357
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for SLA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  SLA_SEG by Gen
## t = 0.13675, df = 179.12, p-value = 0.8914
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -1.530883  1.758867
## sample estimates:
## mean in group G1 mean in group G2
##         17.68480         17.57081
##
## [1] "Shapiro-Wilk test for G1 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.98709, p-value = 0.1714
##
## [1] "Shapiro-Wilk test for G2 d13C_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
```

```
## W = 0.99193, p-value = 0.646
##
## [1] "Levene's test for d13C_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.3468 0.5564
##       283
## [1] "Mann-Whitney U test for d13C_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  d13C_SEG by Gen
## W = 10276, p-value = 0.8099
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for d13C_SEG :"
##
##  Welch Two Sample t-test
##
## data:  d13C_SEG by Gen
## t = 0.4826, df = 281.35, p-value = 0.6298
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -0.1959825  0.3232945
## sample estimates:
## mean in group G1 mean in group G2
##        -29.36454        -29.42820
##
## [1] "Shapiro-Wilk test for G1 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.78099, p-value = 1.344e-12
##
## [1] "Shapiro-Wilk test for G2 msm_all :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.85641, p-value = 8.915e-09
##
## [1] "Levene's test for msm_all :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  1.4254 0.2337
##       234
## [1] "Mann-Whitney U test for msm_all :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  msm_all by Gen
## W = 6132, p-value = 0.1408
## alternative hypothesis: true location shift is not equal to 0
```

```
##
## [1] "t-test for msm_all :"
##
##  Welch Two Sample t-test
##
## data:  msm_all by Gen
## t = -1.1691, df = 219.83, p-value = 0.2436
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -0.755330  0.192848
## sample estimates:
## mean in group G1 mean in group G2
##         3.545736         3.826977
##
## [1] "Shapiro-Wilk test for G1 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.45416, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for G2 LMA_SEG :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.97424, p-value = 0.09012
##
## [1] "Levene's test for LMA_SEG :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0453 0.8318
##       184
## [1] "Mann-Whitney U test for LMA_SEG :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  LMA_SEG by Gen
## W = 4254, p-value = 0.9357
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for LMA_SEG :"
##
##  Welch Two Sample t-test
##
## data:  LMA_SEG by Gen
## t = 0.39456, df = 156.99, p-value = 0.6937
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -0.006078334  0.009112857
## sample estimates:
## mean in group G1 mean in group G2
##       0.06337671       0.06185945
##
```

```
## [1] "Shapiro-Wilk test for G1 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.93746, p-value = 3.476e-07
##
## [1] "Shapiro-Wilk test for G2 fl_duration :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.94171, p-value = 7.61e-06
##
## [1] "Levene's test for fl_duration :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1   1.752 0.1865
##       332
## [1] "Mann-Whitney U test for fl_duration :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  fl_duration by Gen
## W = 14369, p-value = 0.5018
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for fl_duration :"
##
##  Welch Two Sample t-test
##
## data:  fl_duration by Gen
## t = 0.76779, df = 327.35, p-value = 0.4432
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -2.855465  6.511115
## sample estimates:
## mean in group G1 mean in group G2
##         50.54595         48.71812
##
## [1] "Shapiro-Wilk test for G1 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.96875, p-value = 0.002352
##
## [1] "Shapiro-Wilk test for G2 mean_seeds_per_fruit :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.9617, p-value = 0.001835
##
```

```
## [1] "Levene's test for mean_seeds_per_fruit :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.6482 0.4215
##       260
## [1] "Mann-Whitney U test for mean_seeds_per_fruit :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mean_seeds_per_fruit by Gen
## W = 8620.5, p-value = 0.8551
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for mean_seeds_per_fruit :"
##
##  Welch Two Sample t-test
##
## data:  mean_seeds_per_fruit by Gen
## t = 0.0078543, df = 252.22, p-value = 0.9937
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -0.5765515  0.5811686
## sample estimates:
## mean in group G1 mean in group G2
##         3.664109         3.661800
##
## [1] "Shapiro-Wilk test for G1 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.70071, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for G2 est_fecundity :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.61913, p-value < 2.2e-16
##
## [1] "Levene's test for est_fecundity :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0465 0.8294
##       321
## [1] "Mann-Whitney U test for est_fecundity :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  est_fecundity by Gen
## W = 12544, p-value = 0.7101
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fecundity :"
```

```
##
##  Welch Two Sample t-test
##
## data:  est_fecundity by Gen
## t = -0.3123, df = 275.38, p-value = 0.7551
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -13.117804   9.525701
## sample estimates:
## mean in group G1 mean in group G2
##         31.16209         32.95814
##
## [1] "Shapiro-Wilk test for G1 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G1
## W = 0.60755, p-value < 2.2e-16
##
## [1] "Shapiro-Wilk test for G2 est_fitness :"
##
##  Shapiro-Wilk normality test
##
## data:  trait_G2
## W = 0.61576, p-value < 2.2e-16
##
## [1] "Levene's test for est_fitness :"
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1   0.014 0.9059
##       321
## [1] "Mann-Whitney U test for est_fitness :"
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  est_fitness by Gen
## W = 12634, p-value = 0.7927
## alternative hypothesis: true location shift is not equal to 0
##
## [1] "t-test for est_fitness :"
##
##  Welch Two Sample t-test
##
## data:  est_fitness by Gen
## t = 0.05171, df = 305.94, p-value = 0.9588
## alternative hypothesis: true difference in means between group G1 and group G2 is not equal to 0
## 95 percent confidence interval:
##  -6.080374  6.408567
## sample estimates:
## mean in group G1 mean in group G2
##         16.41281         16.24871
```